

Transformer Initialization Geometry

1. Introduction

The Transformer Architecture by (Vaswani et al) has been at the core of deep learning progress. It has shown outstanding performance across various tasks in Natural Language Processing and Computer Vision. The initialization of transformers still remains a highly unexplored topic as still the models are initialized as Gaussian independently and identically distributed weights.

The initialization techniques do not account for Q-K correlation. Previous Works show that randomly initialized weights are a bad condition number as the weights cause imbalance in gradient updates. All the previous works either work with W_q and W_k being i.i.d. or orthogonal, but not as a linear combination of both. What we propose is a formula to study the continuous space of linear combination of random matrix and orthogonal matrix with varying variance of matrices.

By studying the entropy, rank regimes of these single head single layer matrices, we use the knowledge of them to initialize different heads in small Language models in order to prevent heads from going into entropy and rank collapse while maintaining and promoting head diversity.

2. Literature Survey

This is original (jk). Attention is not all you need. R-LoRA. Structured Init, Conditioned init, T-Fix, Unified theory of failures in transformers etc etc. My excel sheety will helpy

3. Experiments

3.1) Experiment 1

In this we train a model (small decoder only). In this we first train a baseline model tracking Rank, Entropy for each head in each layer, and similarity between heads in each layer, and with next layer. The specific details will be filled later.

We initialize $W_q = s \cdot A$, where A is a matrix drawn from iid gaussian of $N(0,1)$

$W_k = s(\alpha A + \beta B + \gamma C)$, where B is orthogonal to A and C is from same distribution of A and $\alpha^2 + \beta^2 + \gamma^2 = 1$

Later we will sweep through intervals of $(s, \alpha, \beta, \gamma)$ and train a single layer and single head transformer model to identify regions in $(s, \alpha, \beta, \gamma)$ which lead to rank collapse and entropy collapse. We will also identify stable regions in $(s, \alpha, \beta, \gamma)$. Standard Init techniques will have $(s, \alpha, \beta, \gamma) = (\sqrt{d_k}, 0, 0, 1)$.

3.2) Experiment 2

After identifying valid regions which lead to stable zones, we will initialize our heads from different zones in our regimes in a multilayer multi head transformers Language model. Our objective in this experiment is to prove that this way to initialization is better than random init in terms for head stability and diversity.

We will train multiple models here with our diverse initialization and random xavier and compare them on basis of generalization gap, train loss, validation loss, #heads collapsed due to rank, #heads collapse due to entropy and #redundant heads.

4. Results

4.1) Results of Experiment 1

We found that this blah blah blah values cause this collapse and blah blah blah values cause this phenomena. Also this region is this and that is that

(few beautiful plots of what is safe and what causes collapse)

4.2) Results of Experiment 2

We found that using our init technique our trained models are better during training time due to (faster convergence/better generalization/ more head diversity/ more head participation)

We trained these models (small, medium, large) and found the results in these models.

5. Limitations

Works of Vidit Gupta have no limitations (jk) we will write any real limitations or just spurt out that this was tested on small models and really large language models will have some internalized phenomena or shit

6. Conclusion

This paper is good and give this paper of the year award at ICLR. Cite this plzz. Love to Chris Manning and Xhang

7. References

[1] Attention is all you need

[2] Attention is not all you need

8. Appendix

A1) Ablation for (s , α , β , γ)